TEACHER QUICK START · 60 MINUTES

ISS GREENHOUSE MODULE

Campaign 1, Case 1 · Gravitropism in Microgravity

Students are learning: Plants normally use gravity as an orientation cue. In microgravity, statoliths do not settle in one consistent direction, so roots lose a reliable “down” signal.

Before class

- Confirm the game loads.
- Print the student sheet double-sided or use fillable HTML mode.
- Keep the evidence fallback ready.
- Provide headphones.

Game access

interstellarharvest.github.io/Space-Sprout-Sleuth/

Select Campaign 1 → Case 1.

Correct diagnosis

Microgravity disrupts gravitropism. Roots need an alternative consistent cue or guidance system.

60-minute plan

Time	Action
0-5	Launch game and materials
5-10	Initial prediction
10-30	Investigate four sources
30-38	Eliminate alternatives
38-48	Diagnose + CER
48-55	Engineering response
55-60	Exit ticket

Evidence students should notice

Source	Essential evidence
Crew	Roots do not grow down; resource adjustments did not solve the pattern.
Sensors	Microgravity is the major unusual condition; nutrients, water, temperature, humidity, and lighting are nominal.
Plants	Dense tangles, poor anchoring, and unusual stem angles.
Logs	Gravitropism/statolith mechanism and possible replacement cues.

Main misconception

Microgravity is not the literal absence of gravity. The ISS and everything inside it are continuously falling around Earth, producing apparent weightlessness.

What to collect

Evidence task · diagnosis and rejected alternative · CER · engineering criterion/constraint · exit ticket

Technical fallback

Use the evidence digest on the final teacher page. Students complete the same academic task without the game.

Teacher line: “Your job is not to click everything. Your job is to determine which evidence best explains the root pattern.”

ISS Greenhouse Module

NGSS SUPPORT

Environmental Science / STEM · Middle school through adult · 60 minutes

Overview

Students investigate a stunted lettuce crop aboard the International Space Station. They compare crew testimony, environmental sensors, direct plant observations, and mission logs to determine why roots form tangled balls despite normal resources. Students explain the gravitropism mechanism and evaluate an engineered alternative cue.

Guiding question

> Why are the lettuce roots tangled and directionless even though water, nutrients, temperature, and lighting appear normal?

Standards alignment

- **MS-ETS1-1:** Supports defining criteria and constraints for a root-guidance solution.
- **MS-ETS1-2:** Supports evaluating competing orientation solutions.
- **SEP:** Constructing Explanations and Designing Solutions; Analyzing and Interpreting Data.
- **DCI:** ETS1.A and ETS1.B, with plant structure/function as supporting science.
- **CCC:** Cause and Effect; Structure and Function.

This lesson supports components of the standards; it does not by itself fulfill an entire performance expectation.

Learning objectives

1. Distinguish observations from explanations across four evidence sources.
2. Identify disrupted gravitropism as the best-supported cause.
3. Explain statolith movement as part of a directional gravity-sensing mechanism.
4. Reject an alternative explanation using evidence.
5. Define a criterion and constraint for an orientation system.

Success criteria

Relevant evidence from at least three sources · correct mechanism · evidence-based alternative rejection · concise CER · directional engineering response.

Vocabulary

gravitropism · microgravity · statocyte · statolith · phototropism

Materials

Chromebooks · headphones · game access · double-sided mission sheet or editable HTML · evidence fallback.

Procedure

ONE PERIOD

Teacher preparation

1. Open the game and confirm Campaign 1, Case 1 loads.
2. Test audio/headphones.
3. Print or share the student sheet.
4. Keep the evidence fallback ready.

0-5 minutes · Setup

Students launch the game and open the mission sheet. State that game score, rank, speed, and optional dialogue are not academically graded.

5-10 minutes · Phenomenon launch

Show the greenhouse image or read the briefing. Ask what evidence would distinguish a resource problem from an orientation problem. Students complete initial thinking; do not confirm the diagnosis.

10-30 minutes · Gameplay investigation

Students visit crew, sensors, plants, and logs. They record only evidence that supports or rejects a cause.

Prompts: Which detail is a symptom? Which normal reading rules out an explanation? Have you used more than one source?

30-38 minutes · Evidence checkpoint

Students rank competing explanations. Confirm that resource readings are normal and that the root pattern is directional and repeated. Do not require optional dialogue completion.

38-48 minutes · Diagnosis and CER

Students submit the game diagnosis after it unlocks, then write a claim, evidence from several sources, and reasoning that links microgravity to a weakened directional cue.

48-55 minutes · Engineering application

Students choose an alternative cue and give one criterion and constraint. Strong responses distinguish a direct root cue from lighting that primarily guides shoots.

55-60 minutes · Exit ticket

Students answer independently. Collect or save the mission file.

Assessment, Access & Accuracy

Checks for understanding

- Separates symptom from cause.
- Uses normal readings to reject alternatives.
- Describes microgravity as apparent weightlessness.
- Links statolith direction to root orientation.
- Proposes a cue that addresses the problem.

Assessment

Formative: prediction, evidence matrix, explanation ranking, teacher questions.

Collected: diagnosis, rejected alternative, CER, engineering response, exit ticket.

Accessibility

- Bullet-point evidence accepted.
- CER frame available but optional.
- Text-to-speech compatible.
- Game evidence may remain open.
- Evidence fallback available.
- No group product required.

Misconceptions

1. **No gravity:** The ISS remains under gravity while continuously falling around Earth.
2. **Roots use only gravity:** Roots also respond to moisture, touch, chemicals, and other cues.
3. **Brighter light fixes roots:** Light is a stronger shoot cue; roots need a more direct guide.
4. **Plants cannot grow in space:** Growth is possible with adapted environmental systems.
5. **Plant pillows recreate gravity:** They manage media and fluids, not a gravity vector.

Science status

Established: gravitropism, statolith involvement, phototropism, plant responses to microgravity.

Plausible engineering: root channels, moisture cues, rotating systems.

Fiction: the specific emergency, crew, SAA, and exact symptoms.

Technical fallback

Distribute the evidence digest. Students complete the same mission sheet; technical failure is not graded.

TEACHER CASE ANALYSIS

Evidence Architecture & Distractors

Evidence map

Source	Essential evidence	Role
Crew	Roots do not grow down; changing resources did not help.	Symptom and failed interventions
Sensors	Microgravity; normal nutrients, water, climate, and lighting.	Strong causal difference and elimination evidence
Plants	Tangled roots, poor anchoring, unusual stems.	Direct biological observation
Logs	Gravitropism/statolith mechanism and possible replacement cues.	Mechanism and design options

Reasoning path

1. Root tangling is the symptom.
2. Normal resources weaken simple resource explanations.
3. Microgravity is the major unusual environmental condition.
4. Logs connect gravity sensing to directional growth.
5. Disrupted gravitropism best explains the pattern.
6. A useful solution supplies another consistent cue.

Distractors

Distractor	Why tempting	Why rejected
Nutrient burn	Pale, stunted plants	Nominal solution; directional tangling is a poor match
Wrong light cycle	Odd stems and pale leaves	16/8 and PAR are nominal; roots are the strongest abnormality
Defective seeds	All plants affected	Pattern repeats with environment and prior plantings
Water problem	Roots seek water	Delivery works; uniform wicking may reduce a cue but does not best explain the case

Boundary note

> The game simplifies a complex plant response. Do not teach that roots are completely random, that statoliths are the only gravity-sensing mechanism, or that plant pillows fully solve orientation.

Instructional identity

Student role: Pattern investigator. **Primary practice:** compare multiple evidence types, eliminate alternatives, explain a mechanism, and define an engineering problem.

Evidence & Alternatives

TEACHER ONLY

Initial thinking

Accept evidence needs such as nutrient readings, water delivery, light settings, root patterns, gravity environment, repeated batches, or mission records.

Evidence matrix

Source	Model evidence	Interpretation
Crew	Roots grow in many directions; nutrient, water, and light adjustments did not fix them.	Supports orientation problem; weakens simple resource causes.
Sensors	Microgravity; other major readings nominal.	Identifies unusual condition and rejects several alternatives.
Plants	Dense tangles, wrapping, weak anchoring, odd stems.	Direct symptom of disrupted directional growth.
Logs	Statolith/gravitropism mechanism and replacement cues.	Explains cause and provides engineering options.

Competing explanations

- **Nutrient damage - weakened:** readings are nominal; tangling is not a strong match.
- **Wrong light cycle - weakened:** schedule/intensity are nominal; lighting does not best explain roots.
- **Defective seeds - weakened:** repeated environment-linked pattern.
- **Microgravity - supported:** matches all four evidence sources and the mechanism.

Mechanism model

Earth gravity → statoliths **settle** → consistent signal → roots grow **with gravity/downward**.

Microgravity → statoliths do not **settle in one direction** → unreliable signal → roots **curve or grow without consistent orientation**.

> Avoid requiring “random.” Roots may still respond to moisture, light, touch, chemicals, and internal growth programs.

Diagnosis model

Microgravity disrupted normal gravitropic orientation. Without a stable settling direction for statoliths, roots lacked a reliable “down” cue and formed tangled growth around the plant pillows.

CER, Design & Exit Ticket

TEACHER ONLY

Rejected alternative model

Nutrient damage is tempting because the plants are stunted and pale, but the sensors show nominal nutrient levels. It also does not explain the roots' consistent directional/tangling problem.

Model CER

CLAIM	The crop problem was caused by disrupted gravitropism in microgravity.
EVIDENCE	Kim reported roots that did not grow downward. Sensors showed microgravity while resources were normal. Plant inspection showed dense tangles and poor anchoring. Logs connected statolith-based sensing to root direction.
REASONING	Gravitropism depends on a consistent gravity direction. In microgravity, statoliths do not settle in one stable direction, so roots lose a reliable orientation signal. This explains the root pattern better than nutrient, lighting, or seed explanations, which are weakened by normal readings and the repeated environmental pattern.

Engineering response

Example design: Directional root channels inside the plant pillow. **Criterion:** most roots should extend into the medium rather than wrapping around the surface. **Constraint:** channels must remain lightweight and permit water and oxygen flow.

Directional light alone is a weaker root solution but can support shoot orientation as part of a combined design.

Exit ticket

Expected: Tangled roots should improve first because channels directly guide root direction and anchoring. Pale leaves may improve later if root structure improves resource access.

Acceptable variation

- Students may use “apparent weightlessness,” “microgravity,” or “reduced reliable gravity cue.”
- Students do not need to describe auxin unless taught separately.
- A rotating chamber, moisture gradient, or combined guide can earn full credit when criteria and constraints are defensible.

SCORING TOOLS

Quick & Analytic Rubrics

Quick classroom rubric

Dimension	Secure	Developing	Beginning
Evidence	Specific, relevant evidence from several sources	Some relevant evidence; important detail/source missing	Vague, copied, or irrelevant evidence
Mechanism	Connects microgravity, statolith direction, and root orientation	Names gravitropism but mechanism is incomplete	Cause stated without mechanism
Reasoning	Connects evidence to claim and rejects an alternative	Partial connection or weak rejection	Evidence listed without explanation
Communication	Accurate vocabulary and complete product	Minor vocabulary/completion gaps	Major gaps impede meaning

Formal analytic rubric

Criterion	4 - Accomplished	3 - Proficient	2 - Developing	1 - Beginning
Evidence selection	Most relevant; symptom/cause distinguished	Relevant with minor omissions	Mixed relevant/irrelevant	Little usable evidence
Source diversity	Integrates 3-4 source types	Uses at least 2 source types	Mostly one source	Sources unclear
Mechanism	Accurate and appropriately qualified	Central mechanism accurate	Important gaps	Unsupported/inaccurate
Alternative evaluation	Specific contradictory evidence	Generally relevant evidence	Weak support	No evaluation
Engineering	Feasible cue, criterion, constraint	Relevant solution + criterion/constraint	Loosely related solution	Does not address orientation
Communication	Clear, concise, complete	Mostly clear/complete	Inconsistent but understandable	Incomplete/difficult

No numeric points appear on the student page by default. Game score, rank, speed, and optional exploration are never academic criteria.

ISS Greenhouse Module

Authoritative sources

1. NASA, *Growing Plants in Space*
   nasa.gov/exploration-research-and-technology/growing-plants-in-space/
2. NASA Ames, *Plant Gravity Perception (SpaceX-13)*
   nasa.gov/ames/space-biosciences/plant-gravity-perception-spacex-13/
3. NASA, *Veggie Plant Growth System Activated on ISS*
   nasa.gov/missions/station/veggie-plant-growth-system-activated-on-international-space-station/
4. NASA Science, *Plant Biology Program - Hardware*
   science.nasa.gov/biological-physical/focus-areas/plant-biology/hardware/
5. Nakamura et al. (2020), *Settling for Less: Do Statoliths Modulate Gravity Perception?*
   pubmed.ncbi.nlm.nih.gov/31963631/
6. NASA OSDR, Arabidopsis Seedlings in Microgravity (OSD-38)
   osdr.nasa.gov/bio/repo/data/studies/OSD-38
7. NGSS MS-ETS1-1 and MS-ETS1-2
   nextgenscience.org

Evidence fallback

Briefing: ISS lettuce is stunted; roots curl around plant pillows instead of anchoring.

Crew: roots do not grow downward; changing resources did not help.

Sensors: microgravity; nutrients, water, temperature, humidity, and lighting are nominal.

Plants: dense tangles, poor anchoring, and angled stems.

Logs: statolith-based gravity sensing normally provides direction; possible compensations include root channels, moisture gradients, shoot lighting, or rotating systems.

Technical notes

- Refresh once if the game does not load.
- Use a current Chromebook Chrome tab.
- The Diagnose screen unlocks after formal clues are collected.
- Do not require every optional conversation branch.
- Move to the fallback within two minutes rather than losing lesson time.

Compatibility

Curriculum prototype v0.2 · Compatible with SSS commit 2a6e8a7 · Prepared 2026-07-22.